

The Certified Min-Entropy Bound for Leg 5

What Liu *et al.* actually prove — and the two things our first draft got wrong

RCS-VRF Subnet — Leg 5 working derivation (rev. 2, reconciled with the paper)

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Status of this revision. An earlier draft derived the Leg-5 bound under an “honest-server” assumption and got the right *formula* but justified it incorrectly. Having now read Liu *et al.* (Nature **640**:343–348, 2025; arXiv:2503.20498) in full, this version replaces that derivation with the paper’s *actual* implemented theorem, and flags exactly what was wrong. The headline number for our honest-server case is unchanged; its justification is now correct and anchored to a published, verified result.

1 What changed from the first draft

Two corrections and one dissolved “open seam”:

1. **The per-round entropy is exact, and it is the *collision* entropy, not von Neumann.** The single-round quantity is the 2-Rényi entropy H_2 of Porter–Thomas, which equals *exactly* $n - 1$ (SM Eq. III.33). Our first draft called $n - 1$ a “conservative von Neumann floor with 0.39 bits of slack.” The value was right; the reasoning was wrong.
2. **There is no EAT / $\sqrt{M}$ / η penalty.** Because the protocol restricts to an i.i.d. adversary, the finite-size bound uses i.i.d. smooth-min-entropy accumulation (Tomamichel–Colbeck–Renner), a clean closed form. The generalized-EAT term we scaffolded and worried about does not appear. That seam is gone.
3. **There is no score→entropy curve $h(\text{XEB})$.** The per-round entropy is a flat $n - 1$; the observed XEB score does *not* modulate it. The score’s entire role is to set $Q_{\min}$, the forced quantum-round count. The real remaining object is $Q_{\min}$ (the adversary analysis), not a min-tradeoff curve.

2 The setup and the one assumption

A client sends M pseudo-random n -qubit circuits, one sample requested per circuit; the server returns $X^M = (X_1, \dots, X_M)$. A random size- m subset is scored, $\text{XEB} = \frac{N}{m} \sum_{i \in V} p_{C_i}(x_i) - 1$ with $N = 2^n$. The protocol accepts iff the average response time is below $t_{\text{threshold}}$ *and* $\text{XEB} \geq \chi$.

Assumption 1 (Honest server — our near-term scope). Every accepted round is sampled faithfully on the QPU. In the paper’s notation this is the special case $Q_{\min} = M$: an honest server runs *all* rounds quantum-mechanically.

The full theorem below holds for *any* server via $Q_{\min}$; Assumption 1 is just the substitution $Q_{\min} \mapsto M$, which is what we can defend today and quote with its label attached.

3 Ingredient 1: the exact per-round entropy (collision entropy)

Lemma 1 (Per-round collision entropy of Porter–Thomas; SM Eq. III.33). *For the Porter–Thomas density $f(p) = Ne^{-Np}$, the conditional 2-Rényi (collision) entropy of one outcome is*

$$H_2(X_i | C_i) = -\log_2 \left(\int_0^\infty N f(p) p^2 dp \right) = -\log_2 \left(N^2 \cdot \frac{2}{N^3} \right) = -\log_2 \frac{2}{N} = n - 1.$$

Why collision entropy, not von Neumann. The i.i.d. accumulation step lower-bounds smooth min-entropy by the *collision* entropy per round, not the Shannon/von Neumann entropy. So $H_2 = n - 1$ is the *correct exact* quantity — not a conservative floor we chose. (For reference, the von Neumann entropy of Porter–Thomas is $n - 0.610$; it plays no role in the bound.)

4 Ingredient 2: i.i.d. accumulation (no EAT, no η)

Because each quantum round is i.i.d. (a stated adversary restriction), the smooth min-entropy of Q i.i.d. quantum rounds obeys the Tomamichel–Colbeck–Renner i.i.d. bound (SM Eq. III.32):

$$H_{\min}^{\varepsilon_s}(X_Q | \tilde{I}_{\text{sn}}) \geq Q \cdot H_2(X_i | C_i) - \log_2 \frac{1}{\varepsilon_s} = Q(n - 1) - \log_2 \frac{1}{\varepsilon_s}.$$

The η seam is gone. No $\sqrt{M}$ second-order term, no generalized-EAT constant to import. The i.i.d. restriction buys the clean closed form. (The asymptotic Aaronson–Hung analysis *does* use EAT, but the *implemented* finite-size bound here does not.)

5 The certified bound, and its numbers

Theorem 1 (Liu *et al.*, Theorem 1 / SM Theorem 3, Eq. III.30). *Given $\varepsilon_s \in (0, \frac{1}{4})$, the protocol either aborts with probability $> 1 - 4\varepsilon_s$, or*

$$H_{\min}^{\varepsilon_s}(X^M | \tilde{I}_{\text{sn}}) \geq Q_{\min}(n - 1) - \log_2 \frac{1}{\varepsilon_s}$$

where $Q_{\min} = \arg \min_Q \{\varepsilon_{\text{adv}}(Q, \chi) \geq 4\varepsilon_s\}$ is the minimum number of quantum rounds any admissible adversary must perform.

Extraction with a quantum-proof strong (Toeplitz) extractor then yields (SM Corollary 7, Eq. III.39)

$$\ell \leq Q_{\min}(n - 1) - 3 \log_2 \frac{1}{\varepsilon_{\text{sou}}} - 2.$$

Verified against the paper. At $n = 56$, $Q_{\min} = 1297$, $\varepsilon_{\text{sou}} = 10^{-6}$, $\varepsilon_s = \varepsilon_{\text{sou}}/4$:

$$H_{\min} \geq 1297 \cdot 55 - \log_2 \frac{1}{2.5 \times 10^{-7}} = 71,335 - 21.9 = \mathbf{71,313} \text{ bits,}$$

matching the paper exactly; and $\ell \leq 71,335 - 3 \log_2 10^6 - 2 = \mathbf{71,273}$ bits, exactly the number they extracted. Our code reproduces both.

Our toy honest-server case. With $Q_{\min} = M$ (Assumption 1), $n = 8$, $M = 1000$, $\varepsilon_s = 2^{-33}$:

$$H_{\min} \geq 1000 \cdot 7 - 33 = 6967 \text{ bits} \quad (\text{vs. the old heuristic } 1659).$$

Unchanged from the first draft — the formula was right; only its justification is now correct. The $\approx 4\times$ gap to the heuristic still decomposes as the heuristic’s two errors (the spurious $\times\text{XEB}$ multiplier and the wrong per-sample constant).

Toy scale, adversarially, is still 0. Run Theorem 1 against a real adversary at $n = 8$ and $Q_{\min} \rightarrow 0$ (a laptop simulates 8-qubit circuits), so the certified entropy is ≈ 0 — by design. The positive 6967 exists *only* under Assumption 1; quote it as “ ≈ 6967 bits, assuming the server did not spoof.”

6 The real remaining object: $Q_{\min}$ (the adversary model)

The score χ and the deadline enter *only* through $Q_{\min}$ (SM Lemma 2):

$$\begin{aligned} \Phi_{\mathcal{A}} &= \min\left(M - Q, \frac{c_{\text{eff}} \mathcal{A}(M t_{\text{threshold}})}{B}\right), & \varepsilon_1 &= \exp\left(-\frac{\delta^2 \Phi_{\mathcal{A}}}{3}\right), \\ L_{\max} &= Q + \Phi_{\mathcal{A}}(1 + \delta), & \varepsilon_2 &= \sum_{l=0}^m \text{Hyp}(l; M, L_{\max}, m) \tilde{\Gamma}^c(m+l, m(\chi+1)), \end{aligned}$$

with $\varepsilon_{\text{adv}}(Q, \chi) = \min_{\delta}(\varepsilon_1 + \varepsilon_2)$ and $Q_{\min} = \min\{Q : \varepsilon_{\text{adv}} \geq 4\varepsilon_s\}$. Here $\mathcal{A}$ is the adversary’s FLOP rate, B the per-circuit simulation cost, c_{eff} the numerical efficiency.

Transcribed, not yet calibrated. Our `compute_q_min` implements this faithfully and is structurally correct (monotone, sensible), but with the paper’s Table-I values it reproduces $Q_{\min}$ only to order of magnitude (~ 1700 vs 1297): the exact conventions for B , c_{eff} , $\mathcal{A}$, and the δ -selection rule live in the authors’ released code (Zenodo 10.5281/zenodo.12952178). It ships with a `calibrated=False` flag and must be reconciled there / verified by Mehdi before any real use. We deliberately did *not* tune parameters to force 1297 — that would be fitting to the answer.

7 What is verified, and what remains

Verified / closed	Remaining
Per-round entropy $H_2 = n - 1$ (exact, Lemma 1).	Calibrate $Q_{\min}$ to the authors’ code (Mehdi).
Certified bound reproduces 71,313 bits.	Real QPU for Leg 4 (Property 1; Colton).
Toeplitz extraction reproduces 71,273 bits; extractor is quantum-proof (closes the quantum-conditional-LHL question).	Timing check in Leg 5 (Property 4).
No η / EAT term needed (i.i.d. accumulation).	At production scale, $Q_{\min}$ becomes the central security argument.

References. M. Liu *et al.*, *Certified randomness using a trapped-ion quantum processor*, Nature **640**:343–348 (2025); arXiv:2503.20498 (Methods + SM Sec. III). S. Aaronson & S.-H. Hung, STOC 2023; arXiv:2303.01625. M. Tomamichel, R. Colbeck, R. Renner, IEEE Trans. Inf. Theory **55**, 5840 (2009).